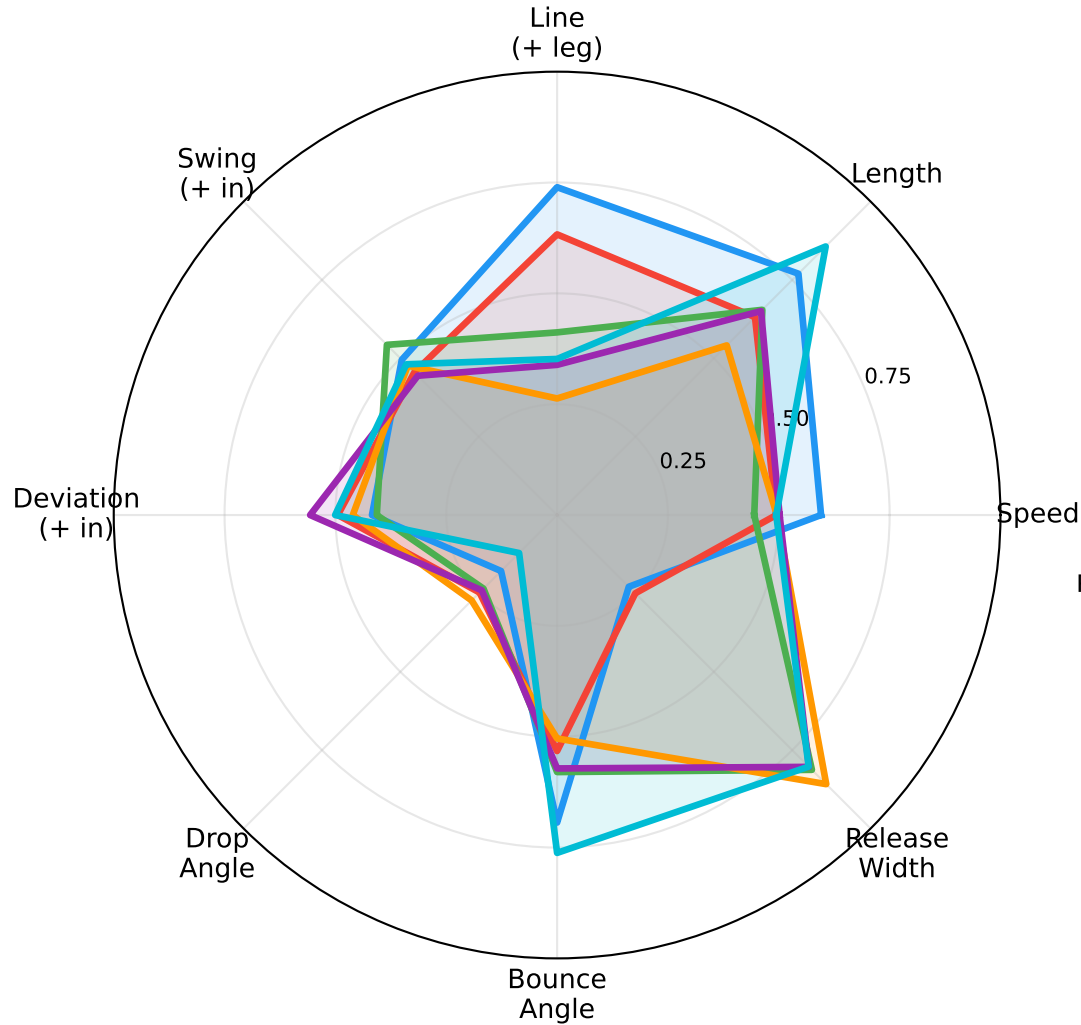


M Morkel — delivery-type clusters

Cluster shapes (normalised)



Cluster centroids (within-bowler z-score)

	Speed (mph)	Length (m)	Line (bat adj)	Swing° (bat adj)	Dev° (bat adj)	Drop Angle°	Bounce Angle°	Release Width
O0 Short n=26, 3.8% wkt	88.3	8.9	0.3	0.1	-0.7	-14.7	10.2	-0.7
O1 Good length n=71, 1.4% wkt	86.2	6.5	0.1	-0.3	-0.1	-12.4	8.1	-0.7
R0 In-swinger n=74, 2.7% wkt	85.1	6.9	-0.3	0.7	-0.8	-12.8	8.7	0.7
R1 Full outside off n=62, 1.6% wkt	86.2	4.9	-0.6	-0.1	-0.4	-11.6	7.7	0.8
R2 Out-swinger n=98, 0.0% wkt	86.3	6.8	-0.4	-0.4	0.3	-12.6	8.6	0.7
R3 Bouncer n=17, 0.0% wkt	86.2	10.4	-0.4	-0.0	-0.1	-16.6	11.2	0.7

z-score

2.0

1.5

1.0

0.5

0.0

-0.5

-1.0

-1.5

-2.0

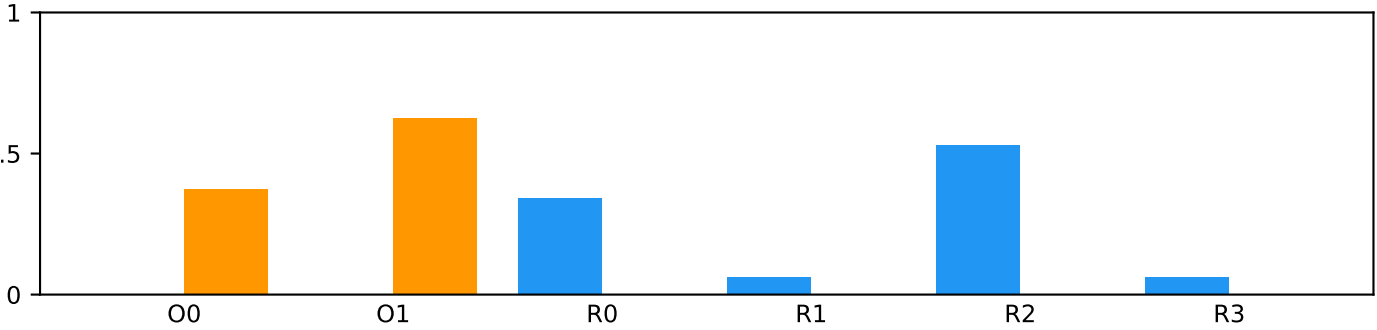
- O0 Short (n=26, 3.8% wkt)
- O1 Good length (n=71, 1.4% wkt)
- R0 In-swinger (n=74, 2.7% wkt)
- R1 Full outside off (n=62, 1.6% wkt)
- R2 Out-swinger (n=98, 0.0% wkt)
- R3 Bouncer (n=17, 0.0% wkt)

M Morkel — per-match strategy mix (page 1 of 2)

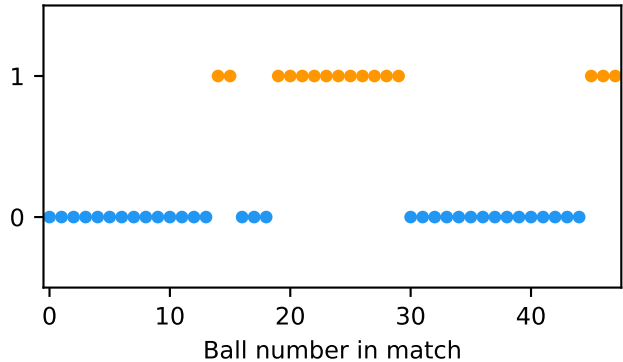
★ = wicket

State 0 State 1

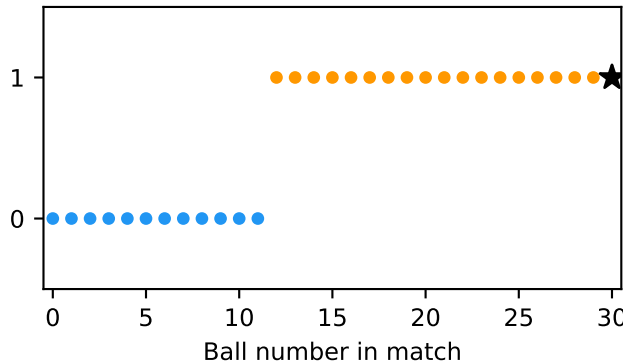
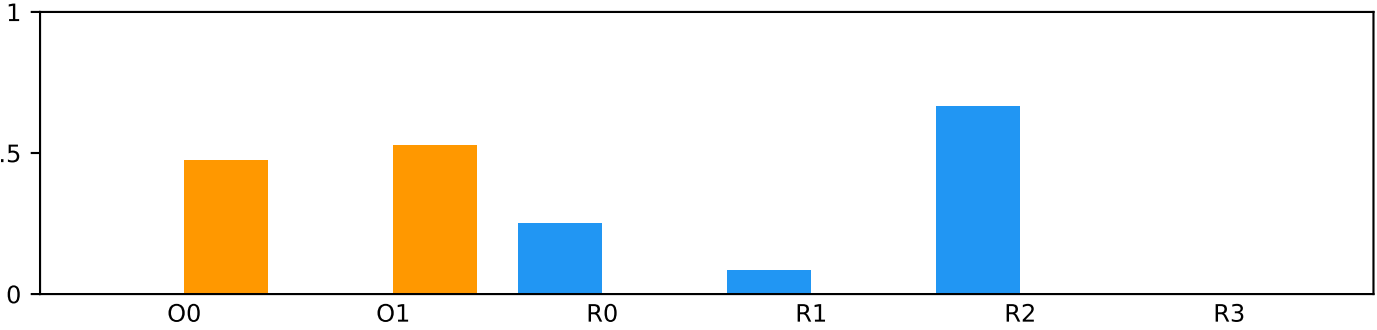
Delivery-type mix per state (see page 1 for cluster names)



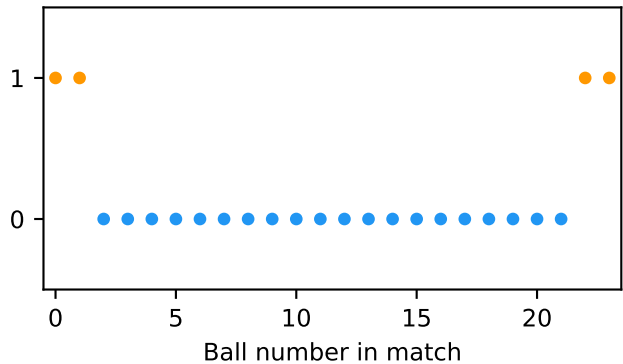
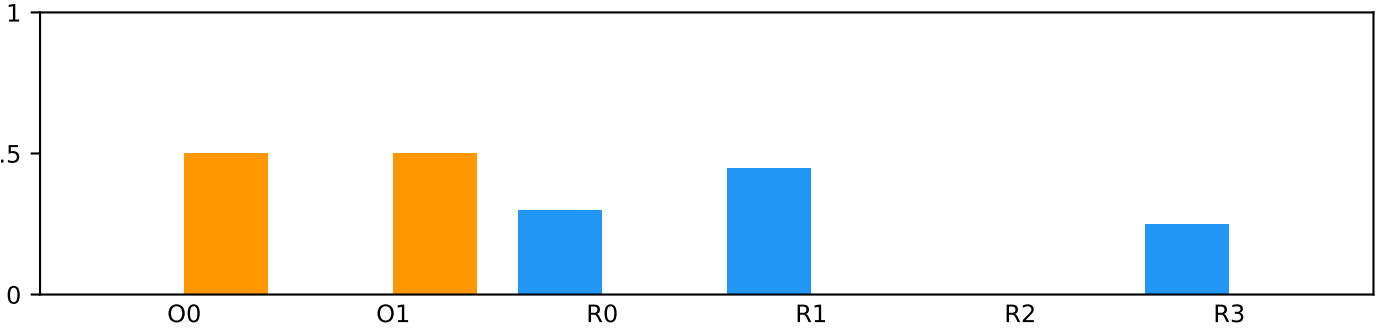
Ball-by-ball state (this match)



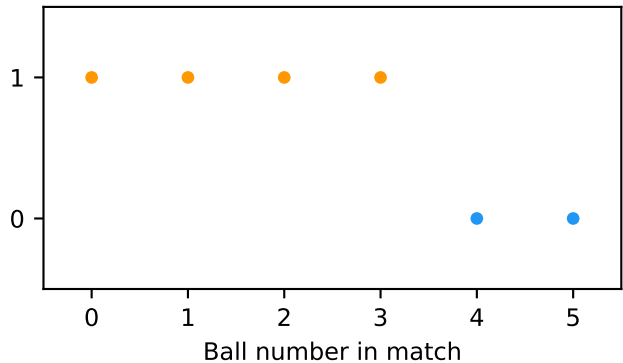
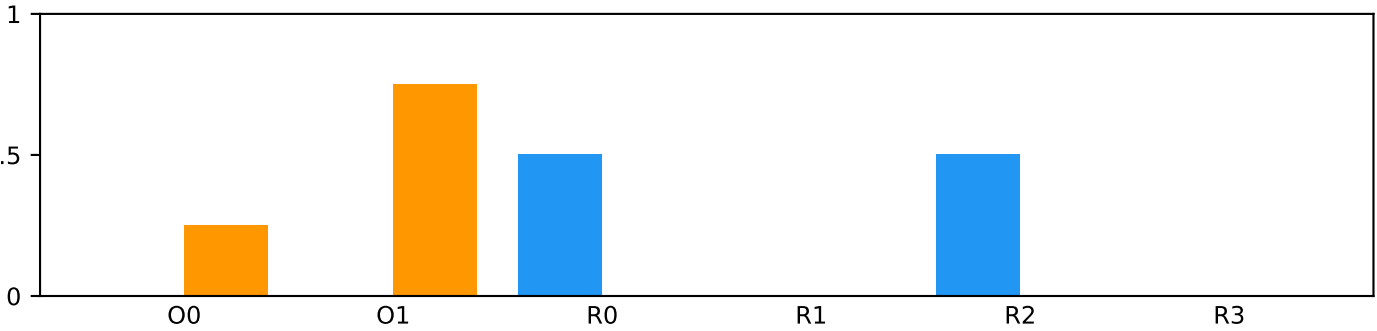
Match 2
2008-10-07
Match ID 17121
Innings 46308



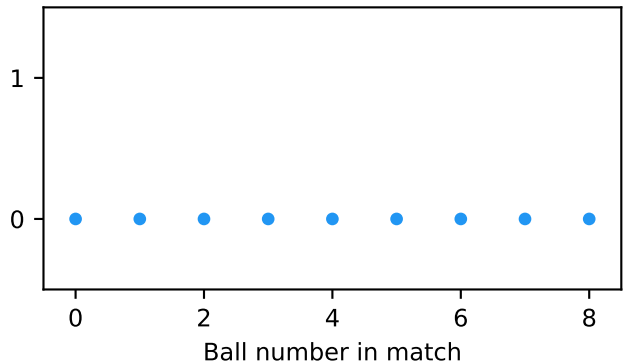
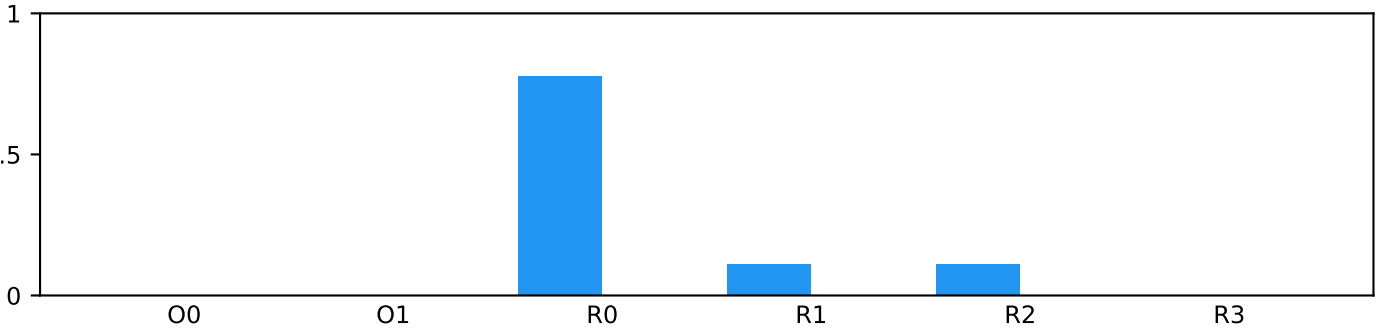
Match 3
2012-02-08
Match ID 27363
Innings 73990, 73992



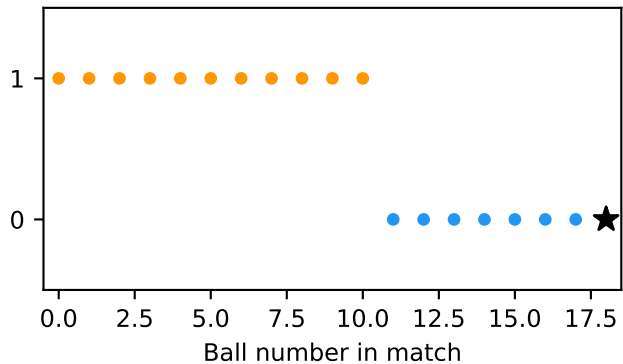
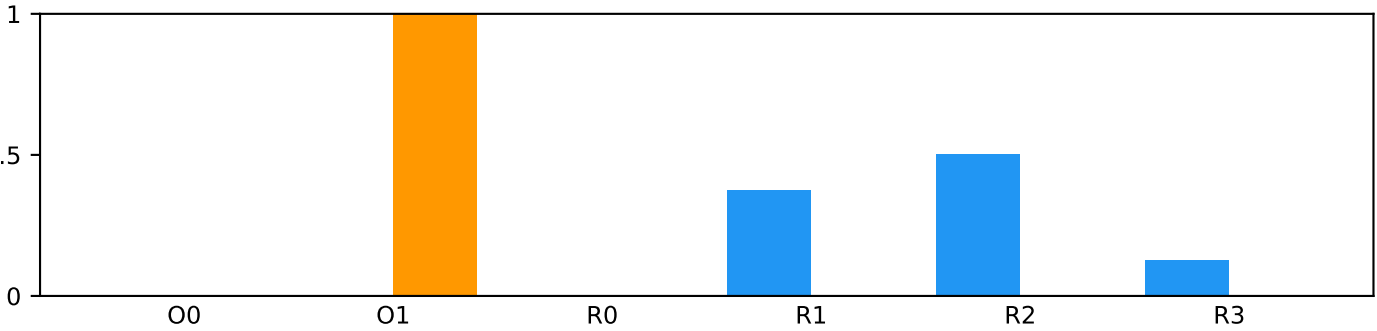
Match 4
2016-02-01
Match ID 34971
Innings 110051



Match 5
2017-04-08
Match ID 38550
Innings 117910



Match 6
2017-06-07
Match ID 38448
Innings 117900



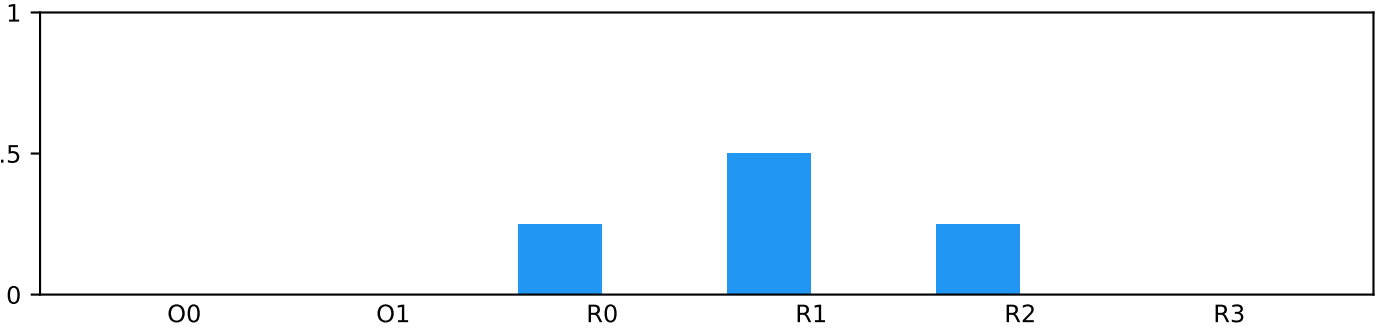
M Morkel — per-match strategy mix (page 2 of 2)

★ = wicket

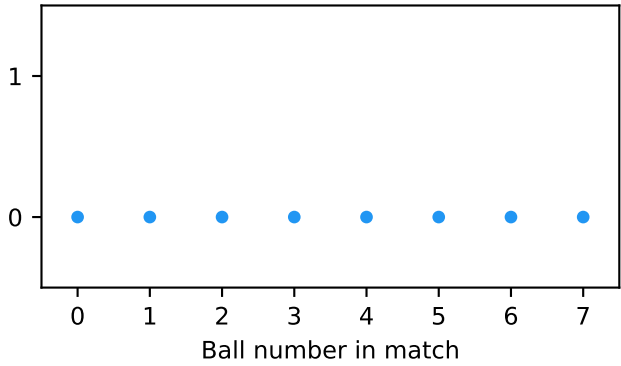
State 0 State 1

Delivery-type mix per state (see page 1 for cluster names)

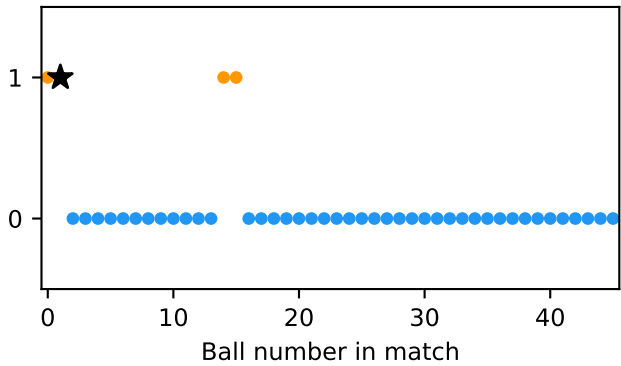
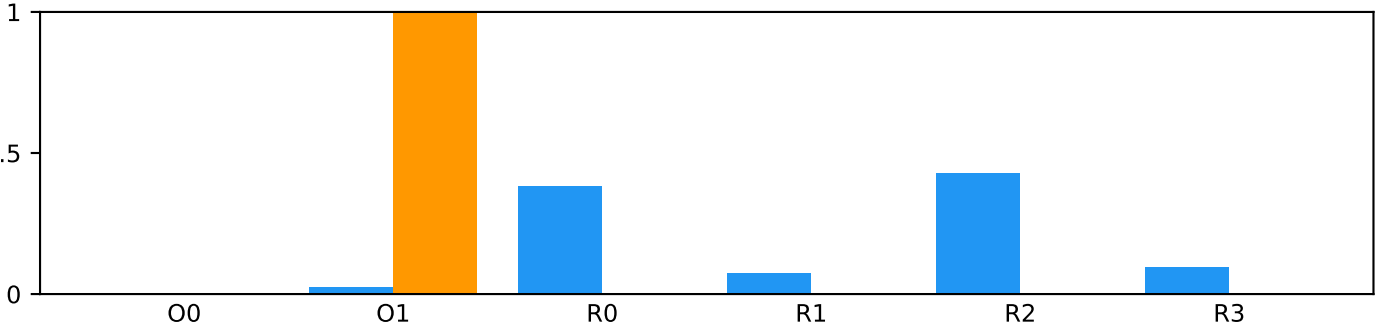
Match 7
2017-07-14
Match ID 38471
Innings 117905



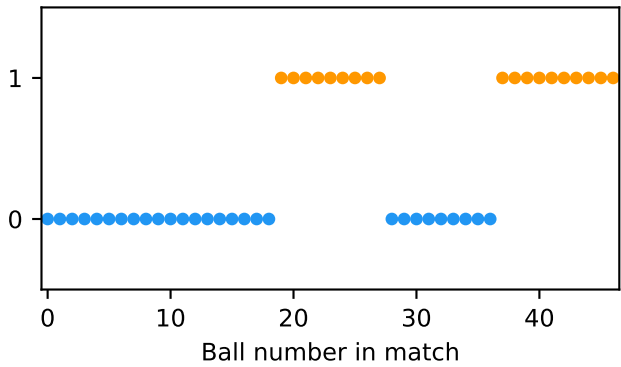
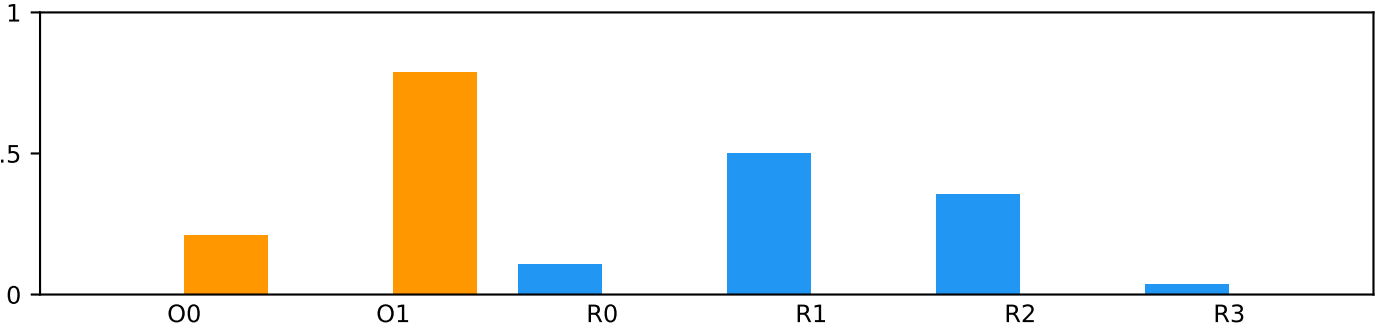
Ball-by-ball state (this match)



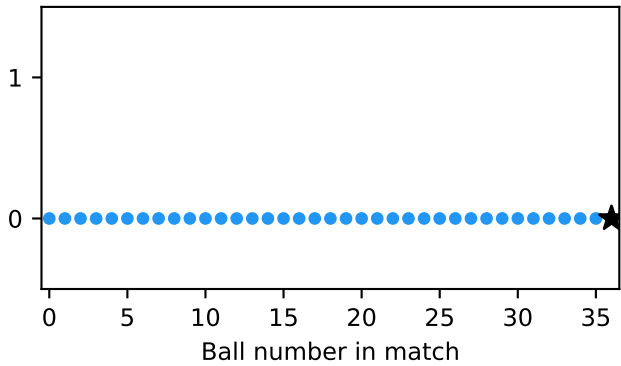
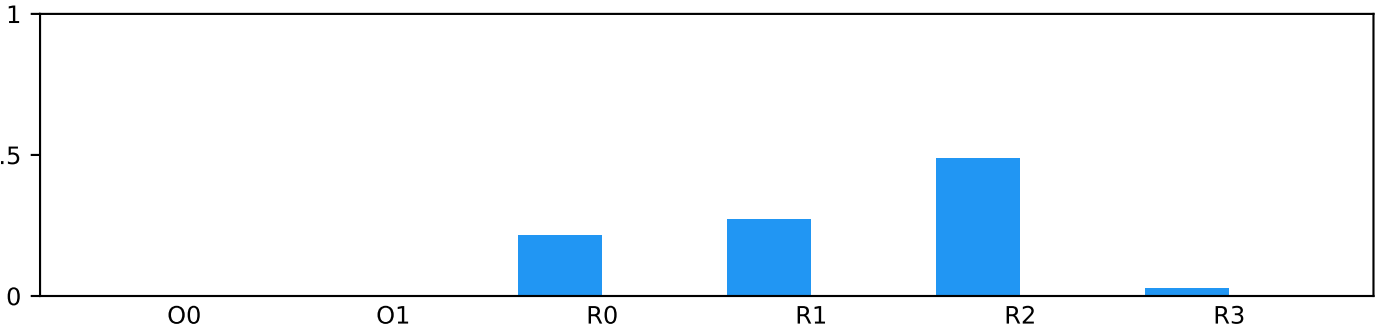
Match 8
2008-07-18
Match ID 17122
Innings 46311, 46313



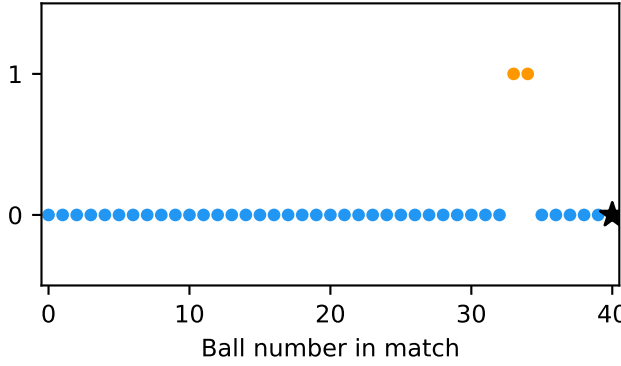
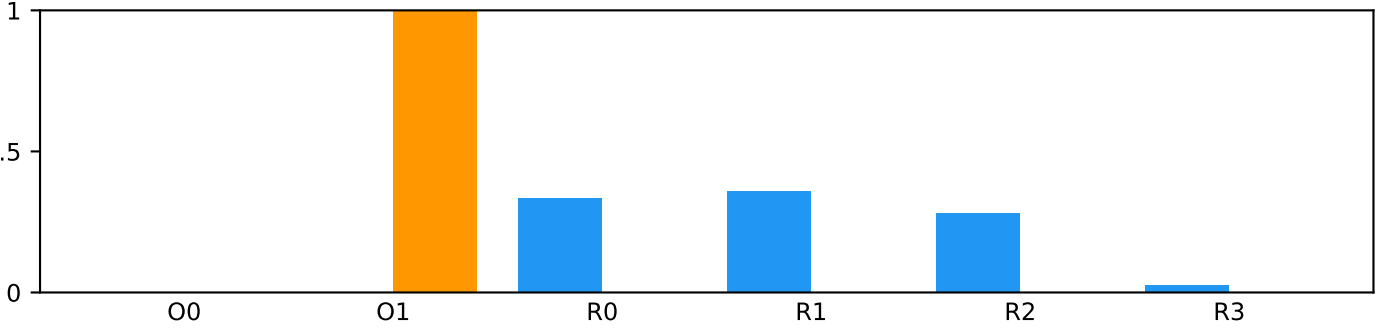
Match 9
2012-07-19
Match ID 27362
Innings 73986



Match 10
2016-01-22
Match ID 35269
Innings 110059



Match 11
2017-07-27
Match ID 38527
Innings 117906



Match 12
2008-07-30
Match ID 17123
Innings 46315

